

Physics-Informed AI-Integrated Digital Twins for Mould Risk Assessment Across Building Retrofit States Using Transfer Learning

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Abstract

Building retrofits are designed to improve energy and thermal performance; however, their influence on surface hygrothermal conditions and associated mould risk has been insufficiently explored, particularly when multiple envelope retrofit measures are implemented concurrently. Within Digital Twin-enabled facility management systems, such physical changes may also compromise the reliability of embedded predictive AI models. To address these challenges, this study advances a physics-informed AI-integrated Digital Twin framework to evaluate retrofit-induced changes in surface-level mould susceptibility and maintain accurate prediction of future mould risk across changing building states. The framework integrates on-site environmental monitoring, Building Information and Energy Modeling (BIM and BEM), VTT-based mould assessment, and Physics-Informed Machine Learning (PIML) within a case-study building in New York. It was then used to assess pre- and post-retrofit mould susceptibility across envelope locations and quantify the individual and combined effects of wall, roof, window, and complete envelope retrofit scenarios through controlled counterfactual simulations. Transfer learning strategies were then evaluated at 1-day-ahead and 1-week-ahead forecasting horizons using limited post-retrofit data to adapt the predictive AI model to the new building state under data-scarce conditions. The completed retrofit reduced mean annualized mould-favorable exposure from 93.2 to 3.1 h per surface per year, a 96.6% reduction, and eliminated exposure at 11 of the 13 investigated locations, although limited residual risk remained at glazing-related junctions. Component-level analysis showed that wall insulation reduced mould-growth hours by more than 93% at most opaque surfaces, whereas glazing replacement reduced growth hours at glazing-associated locations by 77.7 to 88.3%. The PIML models also demonstrated strong transferability across retrofit states. With 20% post-retrofit adaptation data, parameter-based transfer reduced the 1-week-ahead prediction error by 30.0%, while residual transfer reduced the 1-day-ahead prediction error by 12.5% compared with zero-shot prediction. SHAP explainability analysis showed that the transferred models preserved physically meaningful prediction mechanisms, with short-term predictions primarily governed by mould-process physics and longer-term predictions influenced by evolving hygrothermal conditions. The proposed framework therefore advances Digital Twin-based facility management from mould monitoring and forecasting toward retrofit impact assessment, intervention prioritization, and adaptive post-retrofit mould-risk prediction.

Keywords:

Digital Twin; Mould Growth Prediction; Physics-Informed Machine Learning; Building Energy Simulation; Indoor Environmental Monitoring; Building Information Modeling (BIM)

1. Introduction

Dampness and mould continue to pose persistent challenges in residential buildings, with important implications for occupant health, housing quality, maintenance, and regulatory compliance. Surveys across the United Kingdom, China, Australia, and the United States indicate that dampness and mould affect approximately one-quarter to one-half of residential buildings [1-6]. The issue has also received increasing regulatory attention following high-profile cases such as the death of Awaab Ishak in UK social housing [7]. At the same time, the building sector accounts for more than one-third of global energy consumption and nearly 40% of energy- and process-related CO₂ emissions [8], placing substantial pressure on governments to improve the performance of existing building stocks. In the UK, the commitment to achieve net-zero carbon emissions by 2050 has further accelerated the need for energy-efficiency improvements and building retrofit [9]. Consequently, existing buildings are increasingly being upgraded through measures such as improved insulation, glazing, airtightness, heating, and ventilation. While these interventions are primarily implemented to reduce energy demand and improve thermal performance, they can also alter heat and moisture transfer within the building envelope and, consequently, the hygrothermal conditions that govern condensation, dampness, and mould development [10, 11]. However, the success of building retrofit is still assessed primarily in terms of energy and thermal performance, with much less attention given to its effect on mould risk. Improving overall building performance does not necessarily eliminate localized moisture problems, which may persist at vulnerable envelope locations after retrofit. This creates an important gap in post-retrofit assessment, as there is limited evidence showing whether completed interventions actually reduce mould susceptibility across the building envelope or leave residual areas of risk.

Beyond verifying the overall success of retrofit, a second challenge is identifying the contribution of individual interventions. Envelope upgrades may include wall insulation, roof insulation, glazing replacement, or combinations of these measures, yet their specific influence on mould risk is rarely isolated. Conventional pre- and post-retrofit comparisons are often affected by changing environmental and operational conditions, making it difficult to attribute observed improvements to a particular measure. Moreover, even when a retrofit improves overall building performance, untreated or less effective components may continue to create localized areas of mould risk. As a result, facility managers, housing providers, and property developers have limited evidence for determining which retrofit interventions should be prioritized for effective mould-risk reduction. This is particularly important where budget constraints, project schedules, access limitations, and the need to minimize disruption require retrofit measures to be implemented selectively or in phases.

Alongside advances in building retrofit, Digital Twins and Artificial Intelligence (AI) are increasingly being used to support continuous monitoring, performance assessment, and predictive building operation [12-15]. By integrating sensor data, building information, simulation, and predictive analytics, these systems can provide a dynamic representation of building performance and enable earlier identification of emerging risks. For mould management, this creates opportunities for continuous assessment and forecasting beyond conventional periodic inspection. However, predictive models are typically developed for a specific physical and operational state of the building. When retrofit alters the building envelope and its hygrothermal behavior, the relationship between model inputs and mould response may also change, potentially reducing the reliability of the original model. This creates a third gap concerning how predictive capability can be maintained as the building changes. Developing a new model after each major retrofit would require additional monitoring, training, and validation, making this approach impractical for continuously operating Digital Twins. Transfer learning (TL) offers a potential

solution by adapting knowledge from the original building state using limited data from the modified state [16]. However, TL has not yet been applied to mould prediction, leaving unanswered questions about whether an existing model can remain useful immediately after retrofit, how much new data are required for effective adaptation, and whether physically meaningful relationships are retained after transfer.

Accordingly, this study develops an integrated framework to evaluate retrofit effects on mould risk and maintain predictive capability across changing building states. First, changes in surface hygrothermal conditions and mould susceptibility are assessed before and after retrofit. Second, controlled counterfactual simulations are used to isolate the individual effects of wall insulation, roof insulation, glazing replacement, and the complete envelope retrofit under consistent environmental and operational conditions. Third, TL strategies are evaluated to determine whether an existing physics-informed mould-prediction model can be adapted to modified building states using limited target-domain data. Through these analyses, the study addresses three key gaps: the limited verification of mould-risk reduction following retrofit, the lack of evidence on the relative contribution of individual retrofit measures, and the challenge of maintaining predictive mould assessment after the physical building has changed. These contributions extend the role of Digital Twins beyond monitoring and prediction toward retrofit evaluation, intervention prioritization, and adaptive predictive management as buildings evolve.

2. Literature Review

2.1. Building Retrofit, Hygrothermal Performance, and Mould Risk

Building retrofit is widely used to reduce energy demand and improve thermal performance; however, its effect on moisture and mould risk is less consistent. At a broad level, retrofit appears beneficial, but changes to insulation, airtightness, glazing, and ventilation modify heat and moisture transport through the building envelope and can therefore alter the conditions governing condensation and mould development. Lozinsky et al. [17], in a review of residential envelope interventions including insulation, window replacement, air sealing, and combined measures, found that most studies reported reductions in dampness, mould, condensation, and musty odors, often exceeding 30%. However, the review also found insufficient evidence to distinguish the effects of individual measures, leaving uncertainty over which retrofit components are most effective in reducing moisture and mould risk.

More detailed studies demonstrate why this distinction matters, as retrofit solutions achieving similar thermal or energy performance can produce markedly different hygrothermal responses. Park et al. [18], for example, compared a climate-adapted timber wall retrofit with a conventional reinforced-concrete retrofit. Although the two systems achieved almost identical annual energy savings of 25.8% and 26.1%, respectively, the timber system showed better hygrothermal performance over the 10-year simulation, with total water content stabilizing at approximately 3 to 5 kg/m³ compared with 13 to 17 kg/m³ in the conventional retrofit, while also maintaining a lower mould-growth risk. Pihelo and Kalamees [19] reported a similar pattern for a deep external-wall retrofit, where the wall U-value was reduced from approximately 0.9–1.0 to 0.11–0.12 W/m²K, while moisture safety remained strongly dependent on the vapor behavior of the individual layers. Vapor-permeable configurations generally maintained low mould risk, whereas more vapor-resistant configurations could result in moisture accumulation, condensation, and mould growth. These findings highlight that substantial improvements in thermal and energy performance do not necessarily translate into improved moisture safety.

Similar variations in hygrothermal performance are evident in internal wall insulation. Schroderus et al. [20] found that energy retrofit in Finnish multifamily housing reduced purchased energy use by 52%, while the associated mould performance varied considerably between insulation strategies. Thicker

internal insulation became increasingly problematic under future climate conditions, whereas external insulation remained moisture-safe across the configurations examined. Pagoni et al. [21] similarly showed that the performance of internal insulation depended on insulation material, vapor resistance, thickness, indoor moisture load, and façade orientation. Their long-term simulations indicated that increasing insulation thickness could worsen hygrothermal conditions and mould susceptibility. Yuk et al. [22] further demonstrated that even when different insulation systems were designed to achieve the same post-retrofit U-value, their mould and condensation responses differed substantially because of the interaction between the retrofit material and the existing masonry construction. These studies demonstrate that post-retrofit mould performance is shaped not only by the level of thermal improvement achieved, but also by the interaction between the insulation strategy and the existing envelope.

Long-term field evidence further shows that post-retrofit mould risk can remain highly localized, even when the broader wall system performs satisfactorily. Pagoni et al. [23], based on four occupied residential case studies, found that most monitored locations remained below critical mould thresholds, while specific construction details, such as spandrels combining thinner masonry with thicker insulation, exhibited higher mould risk. Indoor moisture conditions also influenced these local outcomes, with elevated moisture loads increasing mould susceptibility behind diffusion-open insulation systems. These findings indicate that acceptable overall wall performance can still coexist with localized areas of elevated mould risk where construction geometry, material configuration, and indoor moisture conditions create less favorable hygrothermal conditions.

Ventilation conditions and building operation also play a major role in shaping post-retrofit moisture and mould performance. Skotnicka-Siepsiak et al. [24] found that thermal upgrading improved internal surface temperatures, but condensation risk persisted when ventilation was insufficient, with an air change rate of approximately 2.0 ACH required to prevent condensation under the investigated conditions. Guan et al. [25] likewise showed that mould susceptibility in highly energy-efficient envelopes depended strongly on operational conditions. Intermittent conditioning increased moisture accumulation and mould risk compared with continuous conditioning, while vapor-control measures that performed well under one operating regime could become less effective under another. This highlights the need to assess retrofit performance in conjunction with ventilation rate, indoor moisture generation, and operational patterns.

Although most of the available evidence concerns wall systems, similar interactions occur in other envelope components. Liu et al. [26]. investigated post-retrofit mould risk in pitched roofs and found that converting a conventional attic into an occupied dormer substantially increased rafter mould susceptibility under low-ventilation conditions, with mould growth index values exceeding 4–5. Higher indoor moisture and ceiling leakage further increased the risk, whereas greater ventilation and the addition of a vapor barrier substantially reduced it. This demonstrates that roof retrofit can also introduce new hygrothermal vulnerabilities when changes in geometry, insulation, moisture transfer, and ventilation are not considered together.

The literature shows that retrofit can improve energy performance while producing variable moisture and mould outcomes across different envelope systems and operating conditions. These outcomes are influenced by retrofit type, material properties, existing construction, ventilation, indoor moisture conditions, climate, and local geometry. Mould risk can also remain concentrated at specific envelope components or construction details even when the overall retrofit performs well. However, most existing studies focus on individual assemblies, selected retrofit strategies, or isolated critical locations. Evidence is still limited on how multiple envelope interventions compare within the same building under

equivalent boundary conditions, how much each intervention contributes to mould-risk reduction, and whether residual hotspots remain after whole-building retrofit. This highlights the need for a spatially resolved, component-level assessment of post-retrofit mould performance.

2.2. Digital Twins for Building Performance and Mould Management

Digital Twins provide a dynamic connection between a physical building and its virtual representation, enabling building conditions and performance to be continuously monitored and analyzed throughout operation [27-30]. A typical Digital Twin comprises the physical asset, its digital counterpart, and the data exchange that links the two. In buildings, this connection can integrate Building Information Modeling (BIM), Internet of Things (IoT) sensor data, simulation models, and operational information within a unified digital environment [31]. As new data become available, the virtual model can be updated to reflect changes in the physical building, supporting continuous performance assessment, predictive analysis, and informed operational decision-making.

The ability to combine physical measurements with virtual building models has supported the application of Digital Twins across a growing range of building-performance domains. Recent studies have demonstrated their use for real-time residential energy management [32], integrated control of HVAC, photovoltaic generation, battery storage, and adaptive shading [33], and building decarbonization through coordinated performance evaluation and energy-system operation [34]. Digital Twin approaches have also been developed for outdoor air-quality monitoring and prediction [35] and for spatially contextualized indoor air-quality management, where BIM, IoT sensing, and AI-based forecasting were integrated to support occupant-specific interventions [13]. Beyond energy and air quality, Digital Twins have been explored as decision-support platforms for managing environmental preferences in shared workspaces [36], maintaining thermal comfort during heatwaves through adaptive closed-loop control [37], and controlling blinds to reduce solar-driven thermal loads while maintaining daylight availability [38]. Across these applications, Digital Twins enable measured data, building information, simulation, and predictive models to operate within a shared digital environment, allowing building performance to be evaluated dynamically and supporting more responsive and context-aware operational decisions.

Extending this broader use of Digital Twins toward more integrated building management, Nie et al. [39] proposed a conceptual framework for architectural heritage that combines structural health, indoor microclimate, and energy performance within a unified system. The framework integrates BIM-based spatial representation, IoT sensing, predictive models, and EnergyPlus simulation to support environmental control and evaluate retrofit alternatives in terms of energy use, carbon emissions, and cost. Mould prevention was incorporated within the microclimate-management component through humidity control and predictive regulation, indicating the potential of Digital Twins to support moisture-sensitive building operation. However, the framework was not implemented or validated using real-building data, and mould risk was not quantified.

Although recent frameworks have begun to incorporate mould prevention within broader microclimate management, Digital Twin applications that explicitly assess and quantify mould risk remain largely unexplored. Existing approaches relevant to continuous mould assessment commonly rely on environmental monitoring and sensor-based assessment [40, 41] or hygrothermal simulation coupled with mould-index modelling [42, 43], typically as separate analytical workflows with limited integration of building geometry, material properties, spatial context, and changing operational conditions. This is a major limitation for mould assessment because mould development is governed by local surface conditions that can differ considerably from room-level air measurements. A Digital Twin can address this limitation by connecting monitored environmental conditions with the physical characteristics and spatial

representation of the building, allowing mould susceptibility to be associated with specific surfaces and envelope components and followed over time.

To address this gap, our previous work [X] developed and validated a physics-informed AI-integrated Digital Twin specifically for mould assessment and prediction in buildings. The framework connected continuous environmental monitoring with a calibrated Building Energy Model (BEM), allowing measured indoor conditions to be extended to surface hygrothermal conditions across monitored and unmonitored envelope locations. These conditions were linked with the Technical Research Centre of Finland (VTT) mould growth model and a physics-informed XGBoost prediction model, creating an integrated workflow for surface-level mould assessment and forecasting. The application demonstrated that the Digital Twin could identify differences in mould susceptibility across the building envelope, including glazing interfaces and lower-resistance wall assemblies that exhibited greater hygrothermal vulnerability.

However, the framework was developed and validated for mould assessment within a single building configuration. Its Digital Twin capability was therefore demonstrated for continuous surface-level assessment under an established physical state, but not for evaluating how mould susceptibility changes when the physical building is modified through retrofit. This leaves a gap in the use of Digital Twins for retrofit assessment, where changes in envelope properties, building components, and local hygrothermal behavior need to be represented and compared across different building states. The present study addresses this gap by extending the established Digital Twin framework to pre- and post-retrofit conditions, enabling consistent evaluation of whole-building and component-level changes in mould susceptibility.

2.3. Transfer Learning for Adaptive Building Performance Prediction

AI has increasingly been applied to mould assessment and prediction using diverse data sources and modelling strategies. Existing approaches include image-based detection of visible mould [44], prediction from housing and facility records [45, 46], identification of fungal contamination from chemical and bioaerosol indicators [40, 47, 48], and prediction of hygrothermal conditions coupled with mould-growth models [41-43]. The latter approaches are more suited to early risk assessment because they estimate the temperature and moisture conditions governing mould development before visible growth occurs. For example, Zhao et al. [42] and Meng et al. [41] used Convolutional Neural Network-Long Short-Term Memory models to predict in-wall temperature and relative humidity before applying the VTT model to assess mould risk, while Aggarwal et al. [43] developed regression models to directly estimate annual mould index from hygrothermal simulations. These studies demonstrate the potential of AI to support predictive mould assessment, but their performance remains dependent on the conditions represented in the data used for model development.

This dependence becomes critical when an AI model is deployed under conditions that differ from its training domain. Changes in building characteristics, climate, occupancy, operation, or envelope properties can alter the relationships between model inputs and the predicted building response, causing a model that performs well in its original domain to lose accuracy when transferred to a new context. Conventional retraining can address this problem, but requires sufficient data from the new domain before the model can be reliably rebuilt. For mould prediction, this creates a practical challenge because mould development is a slow and cumulative process, and obtaining representative target-domain data may require extended monitoring periods. Such data may also be difficult to obtain because of limited instrumentation, access constraints, monitoring costs, or the rarity of conditions associated with elevated mould risk. Consequently, rebuilding a prediction model whenever the building or its operating context

changes can interrupt the continuity expected from a Digital Twin and restrict its ability to adapt to evolving physical conditions.

TL offers a data-efficient alternative by allowing knowledge acquired in a data-rich source domain to be reused in a related target domain where data are limited [16]. Several strategies have been explored in building-performance applications. Instance-based TL identifies and reweights source samples that are most relevant to the target domain [49-51], parameter-based TL transfers learned model parameters and adapts them using a limited amount of target data [52, 53]. Domain-adaptation methods seek to reduce differences between source and target data distributions by learning transferable feature representations [54, 55], and ensemble-based approaches combine multiple transferred models or adaptation strategies to improve robustness across changing conditions [56, 57]. These TL methods have improved building energy and thermal-comfort prediction under cross-building, cross-climate, and data-scarce scenarios. However, applications of TL to mould prediction have not been identified in the existing literature, which presents an opportunity to adapt mould prediction models to changing building conditions.

Model adaptation becomes especially important following building retrofit, as changes to wall, roof, and glazing properties can modify envelope heat and moisture transfer and, in turn, alter the hygrothermal relationships governing mould development. A model trained on pre-retrofit conditions may therefore no longer represent the post-retrofit building accurately.

Our previous work [X] developed and validated a physics-informed XGBoost (PIXGB) model within the Digital Twin framework to predict future surface-level mould conditions. Physics-based information describing the underlying hygrothermal processes was incorporated into the learning framework, enabling mould prediction over multiple forecasting horizons while retaining physically meaningful relationships between environmental conditions and mould development. However, the model was developed and validated using the original building state and its ability to remain accurate following changes to the physical envelope was not investigated. It therefore remains unclear whether a mould-prediction model trained before retrofit can be transferred directly to the modified building, how much its performance deteriorates without adaptation, and whether limited post-retrofit data can restore predictive accuracy without full retraining. The present study addresses this gap by examining the transferability of the established PIXGB model from pre- to post-retrofit building states and evaluating TL strategies for adapting mould prediction under limited target-domain data.

3. Methodology

3.1. Study Design and Framework Advancement

This study builds upon the physics-informed AI-integrated digital twin framework developed and validated in our previous work [X]. As illustrated in Panels A of Figure 1, the framework integrates digital twin-based environmental data acquisition, calibrated BEM for estimating surface hygrothermal conditions, VTT-based mould growth assessment, and physics-informed machine learning for real-time mould evaluation and prediction.

Building on this foundation, the present study extends the established framework to investigate how retrofit interventions influence surface hygrothermal conditions, mould susceptibility, and predictive model performance. As shown in Panels B and C of Figure 1, the framework is applied independently to the pre-retrofit and post-retrofit states of the case study building to quantify the improvements in surface hygrothermal performance and mould susceptibility achieved through retrofit. Controlled counterfactual scenarios are then used to evaluate the stand-alone effects of wall, roof, and glazing upgrades under

consistent environmental and operational conditions. In parallel, TL strategies are developed to adapt the pre-retrofit AI-based mould prediction model to modified building states using limited target-domain data.

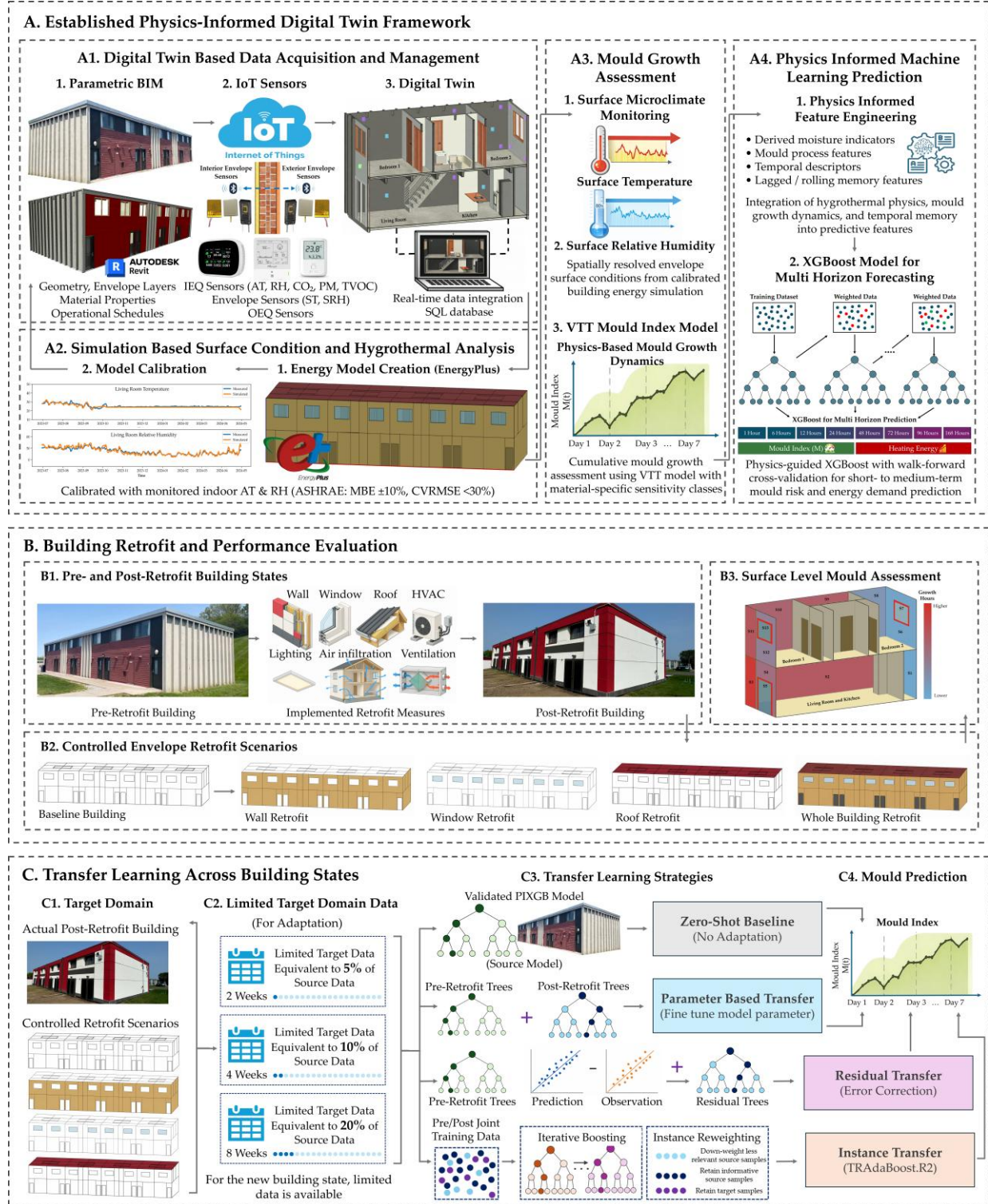


Figure 1. Overall research workflow comprising: (A) the established physics-informed digital twin framework for data acquisition, building energy simulation, mould-growth assessment, and multi-horizon prediction; (B) building retrofit implementation and performance evaluation through controlled envelope retrofit scenarios and surface-level mould assessment; and (C) TL across modified building states using limited target-domain data and zero-shot, parameter-based, residual, and instance-based transfer strategies for mould prediction.

3.2. Case Study Building and Retrofit Intervention

The proposed framework was implemented on an attached residential building located on the Syracuse University campus in New York, USA. Constructed between 1972 and 1973, the building comprises eight single-family two-bedroom duplex apartment units with a total floor area of 608.42 m² and is occupied during the academic year from August to May. One representative apartment (Unit 8) was selected for detailed monitoring and digital twin implementation throughout this study. Figure 2 presents the exterior views of the building before and after retrofit.



Figure 2. Exterior views of the case study building under post-retrofit (top) and pre-retrofit (bottom) conditions.

Prior to retrofit, the building envelope consisted of two external wall assemblies, including a wood-framed wall insulated with fiberglass and a precast concrete wall incorporating Styrofoam insulation. The roof comprised a gypsum ceiling below a precast concrete deck with rigid insulation, while the original fenestration consisted of metal-framed double-pane windows and a double-pane sliding glazed door.

In 2024, the building underwent a comprehensive energy retrofit that modified both the envelope and building services. The envelope upgrades included continuous exterior Expanded Polystyrene (EPS) insulation to the external walls and roof, with replacement of the original glazing with non-metal-framed, triple-pane, low-emissivity, argon-filled windows and a matching low-emissivity sliding glazed door. These interventions increased the thermal resistance of the opaque envelope and reduced heat transfer through the glazing. The resulting post-retrofit envelope properties are summarized in Table 2.

Table 1. Construction layers and thermophysical properties of the post-retrofit building envelope components.

No	Building Envelopes' Layers (Interior to Exterior)	R-Value (m ² ·K/W)	Thickness (cm)	U-Value (W/ m ² ·K)	SHGC
External Wall1	Existing wall + 2" Air Gap + 4.6" Exterior EPS	6.78	29	0.15	-
External Wall2	Existing wall + 2" Air Gap + 5" Exterior EPS	5.89	29	0.17	-
Roof	Existing Roof + 5" Exterior EPS	4.77	27.61	0.21	-
Window	Non-metal frame, triple pane, Low-E, argon	0.88	-	1.14	0.30
Back Door	Non-metal frame, double-pane glazed sliding door, Low-E, argon	0.65	-	1.53	0.40

In addition to the envelope improvements, the building services were upgraded through replacement of the electric resistance baseboard heating system with an air-source heat pump while retaining the original baseboards as auxiliary heating. Additional measures included the installation of an enthalpy energy recovery ventilation system, replacement of the existing lighting with LED fixtures, and improvements to building airtightness. The corresponding operational parameters adopted in the BEM model are summarized in Table 3.

Table 2. Operational and simulation input parameters used in the building energy model.

Parameters	Values
Lighting Power Density	0.17 Watts per Floor Area for each building
Air infiltration	0.15 Air Changes per Hour
Occupancy	400 Area/person (SF/person)
Equipment Power Density	0.5 Watts per Floor Area
Heating System	Air-source heat pump, 4.6 COP at rated condition (electric baseboard retained as backup, 55 °F / 12.8 °C)
Cooling System	Air-source heat pump, 15.2 EER at rated condition
Heating Setpoint	21.1 °C (70 °F)
Cooling Setpoint	25.6 °C (78 °F)
Heating Schedule	15 September to 20 April
Mechanical Ventilation	Energy Recovery Ventilation (enthalpy ERV), continuous 20 cfm bathroom and 25 cfm kitchen, ~70-80% sensible / 60-75% latent effectiveness
Run Period	31/7/2014 to 30/11/2025
Weather File	Actual Meteorological Year (AMY)

3.3. Digital Twin and Building Energy Modelling

3.3.1. Digital Twin Implementation

The digital twin provided the central data and integration environment for linking the physical building with its virtual representation throughout the study. It brought together BIM, environmental monitoring system, centralized SQL database, and physics-based simulation outputs that enables consistent management of measured environmental conditions, simulation outputs, and derived mould-risk indicators, as illustrated in panel A1 of Figure 1.

The selected apartment was monitored before and after retrofit using the same sensing configuration, generating two independent datasets representing the pre-retrofit and post-retrofit building states. Environmental monitoring comprised Indoor Environmental Quality (IEQ), Outdoor Environmental Quality (OEQ), and envelope sensor networks, which continuously measured indoor and outdoor Air Temperature (AT), Air Relative Humidity (ARH), CO₂, Particulate Matter (PM_{2.5} and PM₁₀), Total Volatile Organic Compounds (TVOC), Surface Temperature (ST), Surface Relative Humidity (SRH), and heat flux at selected envelope locations. The monitoring configuration and distribution of the sensors are illustrated in Figure 3(c). Data acquisition, quality-control procedures, and preparation of the datasets followed the methodology reported in the published case-study dataset [58-60].

3.3.2. Building Energy Model Development and Calibration

Building Energy Models (BEMs) were developed for both the pre-retrofit and post-retrofit building states using EnergyPlus, as illustrated in Figure 3(a). Both models were generated from the same BIM geometry while incorporating the envelope constructions, material thermophysical properties, thermal zoning, occupancy schedules, internal loads, infiltration characteristics, and HVAC systems corresponding to their respective building states. Outdoor boundary conditions were represented using Actual Meteorological Year (AMY) weather files generated by integrating the monitored outdoor environmental measurements with a standard weather dataset, in order to reproduce the environmental conditions experienced during each monitoring period.

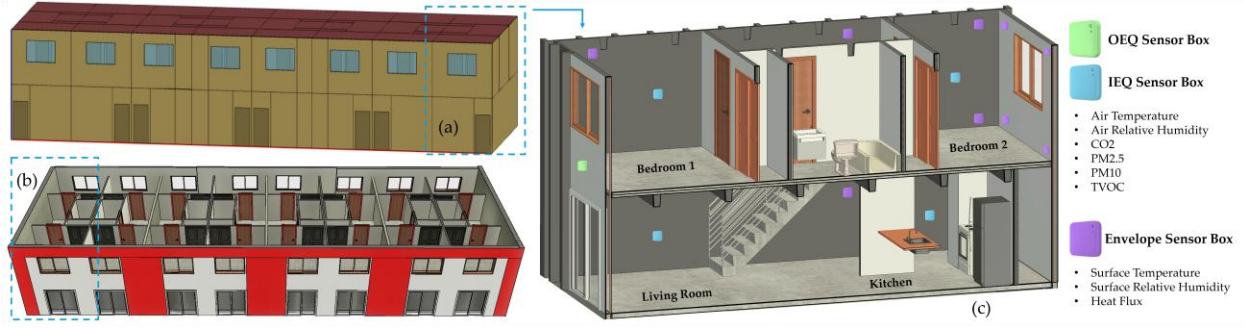


Figure 3. Digital twin modelling and monitoring of the case study building: (a) EnergyPlus building energy model, (b) Revit building information model with the selected apartment highlighted, and (c) BIM representation of the selected apartment showing sensor locations and monitored parameters.

As illustrated in panel A2 of Figure 1, The pre-retrofit BEM previously developed and validated, whereas the post-retrofit BEM incorporated the upgraded envelope assemblies, building services, and operational characteristics described in Section 3.2. Both models were calibrated independently against hourly measurements of indoor AT and ARH obtained from the monitored living room and the east and west bedrooms. Calibration performance was evaluated using the Mean Bias Error (MBE) and the Coefficient of Variation of the Root Mean Square Error (CVRMSE), calculated using Equation (1) and (2), respectively. Following American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Guideline 14, the models were considered satisfactorily calibrated when the hourly MBE remained within $\pm 10\%$ and the CVRMSE remained below 30%.

$$MBE(\%) = \frac{\sum_{i=1}^{N_p} (m_i - s_i)}{\sum_{i=1}^{N_p} m_i} \times 100 \quad (1)$$

$$CV(RMSE)(\%) = \frac{\sqrt{\frac{\sum_{i=1}^{N_p} (m_i - s_i)^2}{N_p}}}{\bar{m}} \times 100 \quad (2)$$

3.4. Surface Hygrothermal and Mould Risk Assessment

3.4.1. Surface Hygrothermal Estimation

The calibrated BEMs were used to estimate hourly ST and SRH at the envelope locations investigated in this study. Although the case study building was instrumented with envelope sensors, direct monitoring of every location is impractical. Physics-based simulation therefore enabled continuous estimation of the surface hygrothermal conditions required for whole-building mould-risk assessment.

Hourly ST and SRH were extracted for the selected wall and glazing-associated surfaces under both the pre-retrofit and post-retrofit building states. The simulated surface conditions were synchronized with the digital twin database to provide a complete spatial representation of the building envelope and subsequently served as the physical inputs to the VTT mould growth model.

3.4.2. VTT-Based Mould Risk Assessment

The estimated surface hygrothermal conditions were converted into mould-risk indicators using the VTT mould growth model, one of the most widely adopted hygrothermal models for evaluating mould development on building materials under transient environmental conditions. Unlike empirical threshold approaches, the VTT model represents mould development as a cumulative biological process governed primarily by ST, SRH, and material sensitivity, allowing both mould growth and decay to evolve continuously in response to changing environmental conditions [61, 62].

The model predicts a time-dependent mould index, $M(t)$, ranging from 0 to 6, where 0 represents no mould growth and progressively larger values indicate increasing biological development until complete surface coverage. Mould growth begins only when the SRH exceeds the temperature-dependent critical humidity threshold, while prolonged unfavorable conditions gradually reduce the accumulated mould index. The maximum attainable mould index is simultaneously constrained by the prevailing hygrothermal conditions to prevent unrealistic unlimited growth. Complete mathematical formulation adopted in this study is available in our previous work [X]. For each investigated surface, the hourly ST and SRH time series were processed sequentially through the VTT model to generate continuous mould-index trajectories throughout the simulation period.

3.5. Physics-Informed Machine Learning Framework

The AI-based prediction component of this study uses the Physics-Informed Extreme Gradient Boosting (PIXGB) as shown in panel A4 of Figure 1. The framework integrates Extreme Gradient Boosting with physics-informed feature engineering to predict future surface mould development by combining monitored environmental conditions with physically informed variables representing the governing hygrothermal and mould-growth processes.

The PIXGB framework was developed using the pre-retrofit building state described in Sections 3.2 to 3.4 and was validated across forecasting horizons ranging from 1 h to 168 h. Its development workflow included chronological data preparation, horizon-specific temporal aggregation, physics-informed feature engineering, feature selection, hyperparameter optimization, and independent model validation. As these procedures have been described in detail [X], they are not repeated here.

In the present study, the previously validated 24 h and 168 h PIXGB models are adopted as the source AI models and form the predictive foundation of the proposed TL framework. These horizons were selected to represent short-term and medium-term mould prediction under different temporal resolutions. The contribution of this study is therefore not the redevelopment of PIXGB, but its systematic adaptation to modified building states using limited target-domain data, enabling the previously learned physical relationships to be transferred and recalibrated following retrofit.

3.6. Retrofit Performance Evaluation Framework

Retrofit performance was evaluated using a two-stage analysis designed to quantify both the overall impact of the completed retrofit and the individual contributions of its principal envelope interventions, as demonstrated in panel B of Figure 1. The first stage compared pre-retrofit and post-retrofit building states, each evaluated using its corresponding monitoring period and AMY weather file, to assess the overall changes in surface hygrothermal behavior and mould risk following retrofit. Because both building states represent the complete retrofit configuration of their respective periods, this comparison reflects the combined effect of all implemented interventions.

To isolate the contribution of individual envelope measures, a second stage employed controlled counterfactual simulations under a common set of environmental and operational conditions. A reference model was established by combining the original pre-retrofit envelope with the post-retrofit building services, operational characteristics, simulation period, and AMY weather file. Individual retrofit scenarios were then generated by introducing wall insulation, roof insulation, glazing replacement, and the complete envelope retrofit while keeping all remaining model inputs unchanged. Consequently, the differences between each retrofit scenario and the reference model quantify the stand-alone effect of the corresponding envelope intervention. Retrofit performance was evaluated using average ST, average SRH, maximum SRH, equivalent annual mould-growth hours, peak mould index, and maximum attainable mould index.

3.7. Transfer Learning Across Building States

To extend the applicability of the validated PIXGB models to modified building states without requiring complete model redevelopment, TL was employed between the pre-retrofit and post-retrofit building domains. The monitored pre-retrofit building dataset was defined as the source domain, while the post-retrofit datasets corresponding to the reference configuration and the individual retrofit scenarios constituted the target domains. Although both domains share the same prediction task, target variable, and physics-informed feature space, the retrofit interventions alter the underlying heat and moisture transfer within the building envelope, resulting in domain shifts in surface hygrothermal conditions and mould-growth behavior.

The TL problem was formulated as a supervised domain adaptation task in which the predictive knowledge captured by the source PIXGB models was progressively transferred to the target domains using limited labelled target-domain data. To prevent developing independent prediction models for each retrofit configuration, TL exploits the physical relationships embedded within the source models while adapting them to the modified environmental conditions introduced by retrofit. The framework was implemented independently for the 24 h and 168 h forecasting horizons, and four adaptation strategies were investigated: zero-shot transfer, in which the source model was applied directly without adaptation; parameter-based transfer through continued boosting; model-based transfer through residual correction; and instance-based transfer using Two-Stage Transfer AdaBoost for Regression (TrAdaBoost.R2) as illustrated in panel C of Figure 1.

Target-domain datasets were generated for the reference configuration and each retrofit scenario using the same predictor definitions and feature representation adopted by the source PIXGB models, enabling direct transfer without modification of the input space. TL experiments were restricted to surface-scenario combinations exhibiting measurable mould development. Combinations with negligible mould growth were excluded because the target response remained effectively constant throughout the simulation period, providing insufficient information for meaningful model adaptation and performance evaluation.

3.7.1. Implementation of the Transfer Learning Framework

Zero-shot transfer served as the baseline by directly applying the source PIXGB model to the target domain without adaptation, parameter updating, or additional training, as illustrated in Figure 4. The resulting predictions therefore quantify the performance obtained when the pre-retrofit model is used directly under modified building conditions.

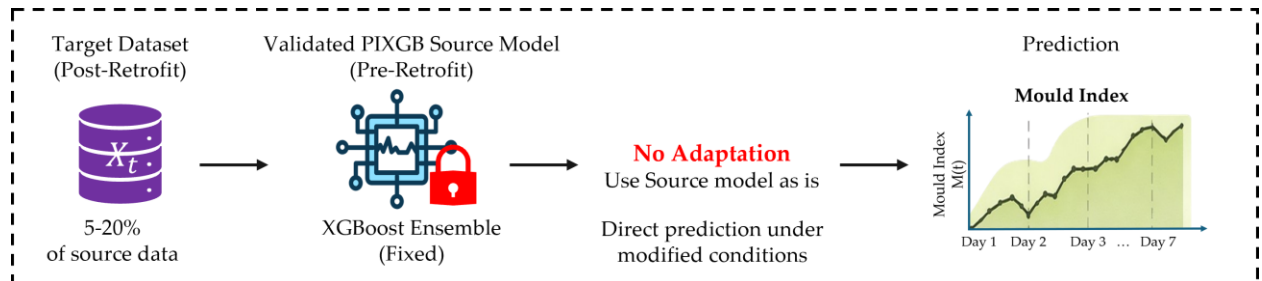


Figure 4. Zero-shot transfer: direct application of the pre-retrofit PIXGB model to the target domain without adaptation.

Parameter-based transfer, illustrated in Figure 5, was implemented through continued XGBoost boosting, where the validated source ensemble was supplied as the initial model and extended using target-domain data. Instead of modifying the existing trees, new boosting trees were appended to progressively reduce the target-domain prediction error. Hyperparameter optimization was performed over four

candidate configurations comprising 40, 80, 120, and 160 additional trees, with learning rates ranging from 0.02 to 0.05 and maximum tree depths between 2 and 4.

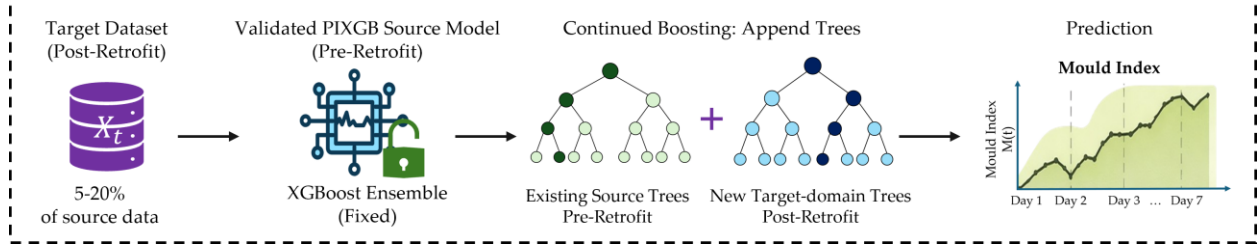


Figure 5. Parameter-based transfer: continued XGBoost boosting by appending new trees trained on target-domain data to the fixed source ensemble.

Residual transfer, shown in Figure 6, retained the source model unchanged while training a secondary XGBoost model to predict the residual error between the source-model predictions and the target observations. The final prediction was obtained by adding the estimated residual correction to the original PIXGB prediction, allowing adaptation to the target domain without modifying the source ensemble. Hyperparameter optimization considered four residual-model configurations with 60 to 200 trees, learning rates between 0.03 and 0.05, and maximum tree depths ranging from 2 to 4.

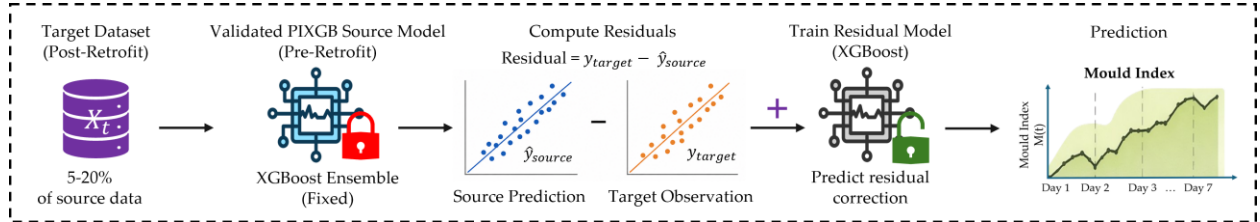


Figure 6. Residual transfer: correction of the fixed source-model predictions using a secondary XGBoost model trained on target-domain residual errors.

Instance-based transfer, presented in Figure 7, was implemented using Two-Stage TrAdaBoost.R2 with a Decision Tree Regressor as the base learner. Source and target samples were jointly used during training, while iterative instance reweighting progressively reduced the influence of source observations that were no longer representative of the target domain and retained those that remained informative. Hyperparameter optimization considered 20 and 30 boosting estimators with maximum base-tree depths of 3 and 4, respectively.

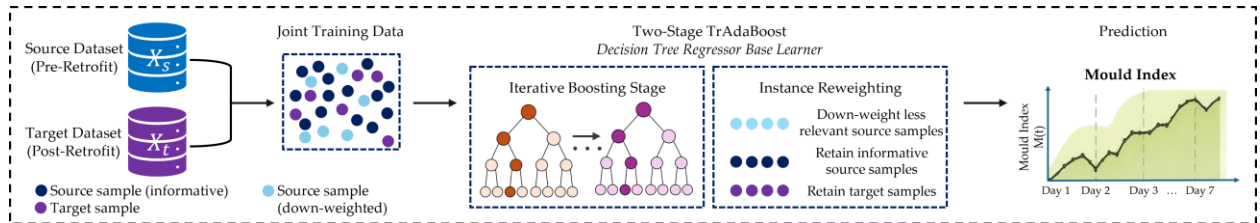


Figure 7. Instance-based transfer: TrAdaBoost.R2 using joint source-target training and iterative instance reweighting.

3.7.2. Experimental Design and Performance Evaluation

A central objective of the TL experiment was to determine the amount of newly available target-domain data required to improve predictive performance beyond zero-shot transfer. Adaptation subsets representing approximately 5%, 10%, and 20% of the observations used to train the original source PIXGB models were therefore evaluated. For the 24 h forecasting horizon, these levels corresponded to the first 15, 30, and 61 daily observations, respectively, while for the 168 h horizon, they corresponded to the first 2, 4, and 8 weekly observations. At each adaptation level, all transfer strategies used the same chronological

adaptation subset and were evaluated on the same remaining post-adaptation observations, ensuring direct comparability between methods under equivalent data-availability conditions.

Prediction performance was assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). RMSE was adopted as the primary comparison metric because it expresses prediction error directly in mould-index units and assigns greater weight to larger deviations. The metrics were first calculated separately for each surface-scenario combination and then averaged across all selected cases.

3.7.3. Model Interpretation Using SHAP

SHapley Additive exPlanations (SHAP) were used to assess whether TL altered the physical relationships learned by the original PIXGB models. The zero-shot source models were compared with the best-performing adapted models at the 20% adaptation level using the independent target-domain test datasets. TreeExplainer was applied to the complete continued ensemble for parameter-based transfer, while residual-transfer SHAP values accounted for the additive contributions of the frozen source and residual correction models. Predictors were grouped into five physics-informed categories: temporal features, hygrothermal conditions, mould-process physics, IEQ, and OEQ. Group importance was calculated as the mean absolute aggregated SHAP value across the test observations, enabling comparison of the physical prediction mechanisms before and after adaptation.

4. Results and Discussion

4.1. Building Energy Model Calibration

The post-retrofit BEM was calibrated at the thermal-zone level against hourly measurements of indoor AT and ARH. The resulting agreement was strong, with AT yielding an average MBE of -1.59% and CVRMSE of 6.42% , while the corresponding ARH values were 3.14% and 15.22% . All metrics remained well within the ASHRAE hourly calibration criteria of $\pm 10\%$ for MBE and 30% for CVRMSE, indicating low systematic bias and acceptable residual variability across the monitored zones. The comparison presented in Figure 8 for the living room confirms that the simulated AT and ARH profiles reproduce the temporal behavior of the measured data over the monitoring period. The calibrated model was therefore considered sufficiently reliable for estimating ST and SRH at unmonitored envelope locations and for supporting the subsequent hygrothermal and mould-risk analyses.

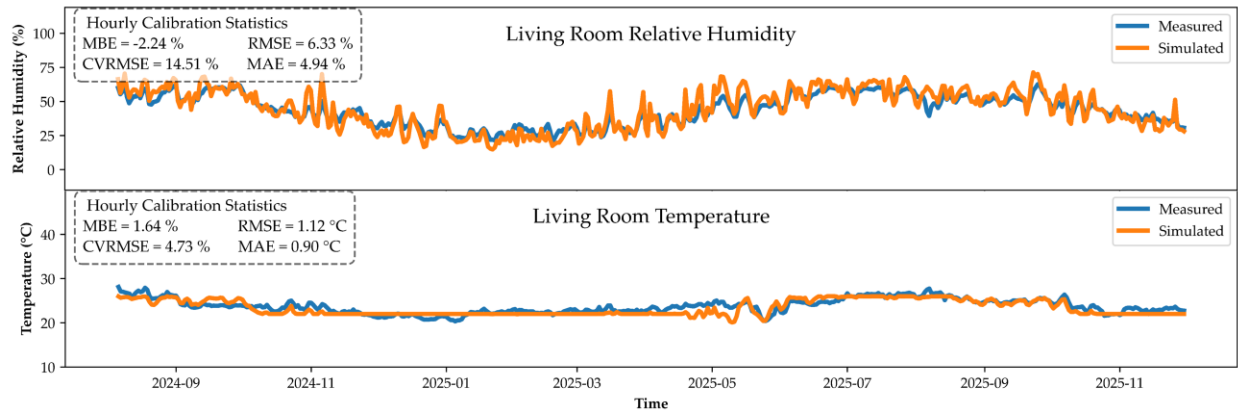


Figure 8. Hourly comparison of measured and simulated indoor AT and ARH in the post-retrofit building.

4.2. Spatial Analysis of Mould Risk Across the Envelope Surfaces Before and After Retrofit

The effect of the whole-building retrofit was evaluated by comparing surface-level mould risk across 13 envelope locations before and after retrofit. Figure 9 presents the spatial distributions of

equivalent annual growth hours and the maximum attainable mould index, $M_{\max}$, while Table 3 reports the corresponding ST, SRH, and mould indicators. To account for the different monitoring durations, growth hours were normalized to equivalent hours per year.

The pre-retrofit risk pattern partly reflected the marked difference between the two external wall assemblies. EW1 had an R-value of 4.13 m²·K/W, whereas EW2 provided only 0.80 m²·K/W. Under the same room conditions, the lower resistance of EW2 produced colder internal surfaces and greater SRH, as demonstrated by the higher exposure at S2 and S3 than at S1 and S4 in the living room. A similar contrast occurred between the façades adjoining Bedrooms 1 and 2, with greater exposure concentrated on the thermally weaker side of the unit. The whole-building retrofit reduced mean annualized mould-favorable exposure across all the surfaces from 93.2 to 3.1 h per surface per year, representing a reduction of 96.6%. Exposure was eliminated at 11 of the 13 locations and the spatial pattern therefore changed from distributed envelope-wide exposure to limited residual risk at two glazing-related locations of S5 and S13.

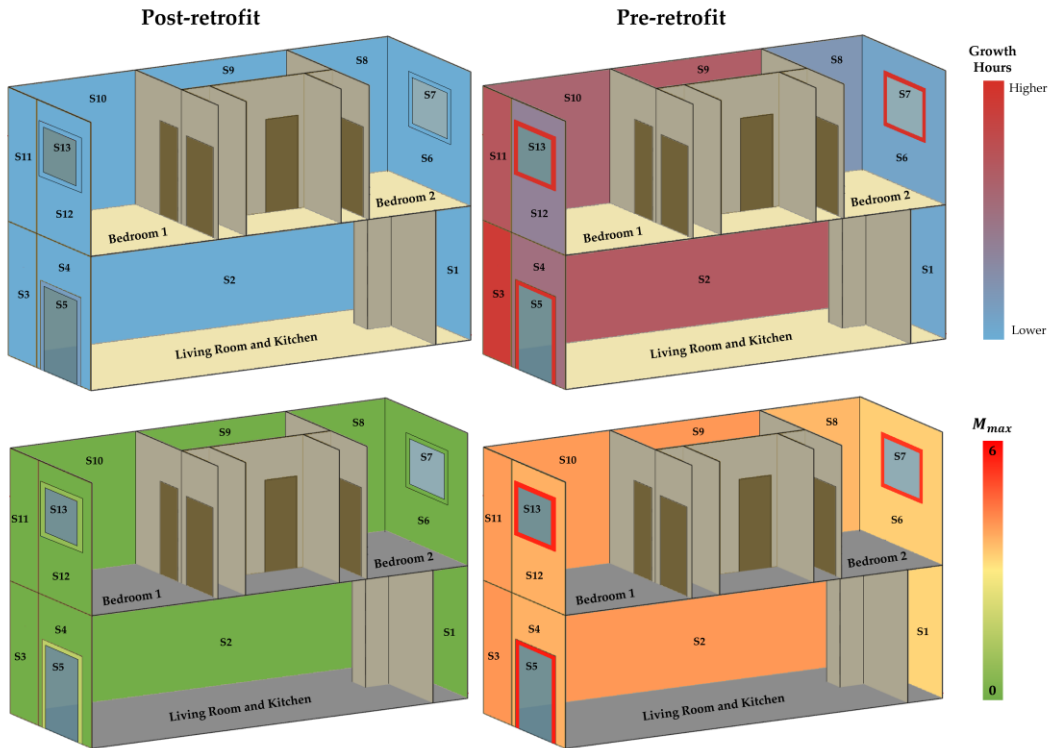


Figure 9. Spatial comparison of mould risk across the building envelope surfaces before and after retrofit.

The reduction across the opaque envelope is consistent with the improved thermal continuity produced by the exterior insulation. Following retrofit, the R-values of EW1 and EW2 increased to 6.78 and 5.89 m²·K/W, respectively, substantially narrowing the performance difference between the two wall systems. At S3, for example, average ST increased from 21.5 to 22.7 °C and maximum SRH decreased from 86.6% to 79.2%. Its annualized growth exposure consequently declined from 39.6 h/year to zero. Similar elimination of mould-favorable exposure occurred at other wall surfaces.

Table 3. Comparison of surface hygrothermal conditions and mould growth indicators before and after retrofit. Values are presented as pre-retrofit / post-retrofit.

Surface	Average ST (°C)	Average SRH (%)	Maximum SRH (%)	Growth Hours/Year	Mould Index	$M_{\max}$
S1	24.0 / 23.1	38.7 / 40.6	80.9 / 73.7	2.4 / 0	0.007 / 0	0.58 / 0
S2	21.8 / 22.7	43.0 / 45.9	86.4 / 79.1	28.8 / 0	0.123 / 0	2.12 / 0
S3	21.5 / 22.7	43.9 / 46.1	86.6 / 79.2	39.6 / 0	0.130 / 0	2.18 / 0

S4	22.8 / 22.7	41.0 / 46.0	84.1 / 78.6	21.6 / 0	0.077 / 0	1.48 / 0
S5	17.3 / 20.7	55.7 / 51.1	100 / 84.7	489.9 / 37.9	1.900 / 0.10	5.32 / 1.65
S6	25.0 / 23.4	36.2 / 43.9	81.5 / 74.7	3.6 / 0	0.011 / 0	0.76 / 0
S7	19.7 / 23.1	49.8 / 45.1	96.8 / 79.2	160.9 / 0	0.729 / 0	4.63 / 0
S8	23.6 / 23.3	38.8 / 43.9	83.4 / 74.9	8.4 / 0	0.030 / 0	1.30 / 0
S9	21.8 / 22.7	43.2 / 46.6	86.5 / 79.0	27.6 / 0	0.150 / 0	2.13 / 0
S10	22.0 / 22.4	42.4 / 46.4	86.1 / 76.8	25.2 / 0	0.095 / 0	2.05 / 0
S11	21.8 / 22.3	43.0 / 46.6	86.0 / 77.0	30.0 / 0	0.098 / 0	2.02 / 0
S12	23.1 / 22.4	40.1 / 46.5	84.2 / 76.2	15.6 / 0	0.045 / 0	1.52 / 0
S13	17.5 / 21.1	54.9 / 49.7	99.0 / 81.4	357.8 / 3.0	1.380 / 0.01	5.11 / 0.72

The highest pre-retrofit risk was concentrated around the windows and glazed doors, where low thermal resistance created cold conditions at adjacent areas. Before retrofit, S5 and S13 recorded average ST values of 17.3 and 17.5 °C, with maximum SRH reaching 100% and 99.0%, respectively. S7 showed a similar pattern, with an average ST of 19.7 °C and a maximum SRH of 96.8%, indicating repeated near-saturation and condensation events around the original glazing assemblies. However, glass is classified as a resistant material in the VTT model, so sustained mould growth on the glazing surface itself is unlikely. The greater concern is the transfer of moisture to adjacent, more mould-susceptible materials, such as wall finishes, window reveals, sealants, and framing elements. At these locations, glazing heat loss and thermal bridging lower the surrounding ST, while repeated condensation and localized moisture accumulation create conditions that are more favorable for mould development.

Glazing replacement substantially reduced the severity of the original hotspots. At S5, average ST increased from 17.3 to 20.7 °C, while maximum SRH decreased from 100% to 84.7%. Annualized exposure consequently fell from 489.9 to 37.9 h/year, and the mould index decreased from 1.90 to 0.10. Exposure was completely eliminated at S7 and reduced from 357.8 to 3.0 h/year at S13. Nevertheless, the limited residual risk at S5 and S13 indicates that glazing junctions remained the most sensitive locations after retrofit.

4.3. Relative Contributions of Individual Envelope Retrofit Measures

To quantify the contribution of each envelope component independently, wall, roof, glazing, and full-retrofit scenarios were compared against a common counterfactual baseline generated using the calibrated post-retrofit BEM. Consequently, all scenarios were evaluated under identical weather conditions, occupancy schedules, HVAC operation, and simulation period, allowing the influence of each envelope measure to be isolated from non-envelope factors.

Figure 10 demonstrates that the effectiveness of each retrofit measure was highly dependent on the affected surface type. Wall retrofitting delivered the largest improvements across the opaque envelope, where mould growth hours were reduced by more than 93% at most wall surfaces and completely eliminated at S1, S6, and S8. However, the glazing-associated hotspots (S5, S7, and S13) remained only partially improved, with reductions in mould growth hours of 31.9%, 37.5%, and 36.7%, respectively, while no reduction in the maximum attainable mould index ($M_{\max}$) was observed because these locations continued to reach saturation. In contrast, glazing replacement produced the opposite response. Although average SRH decreased by only 16 to 24% at the window surfaces, mould growth hours were reduced by 77.7 to 88.3%, the mould index by 85.3 to 93.9%, and $M_{\max}$ by 34.7 to 48.7%. This disproportionate response reflects the threshold-dependent nature of mould growth, whereby relatively modest improvements in local hygrothermal conditions substantially reduce the duration of conditions favourable for biological growth. Roof retrofitting produced only marginal changes across almost all indicators, confirming that heat transfer through the roof was not the dominant driver of mould formation in this dwelling.

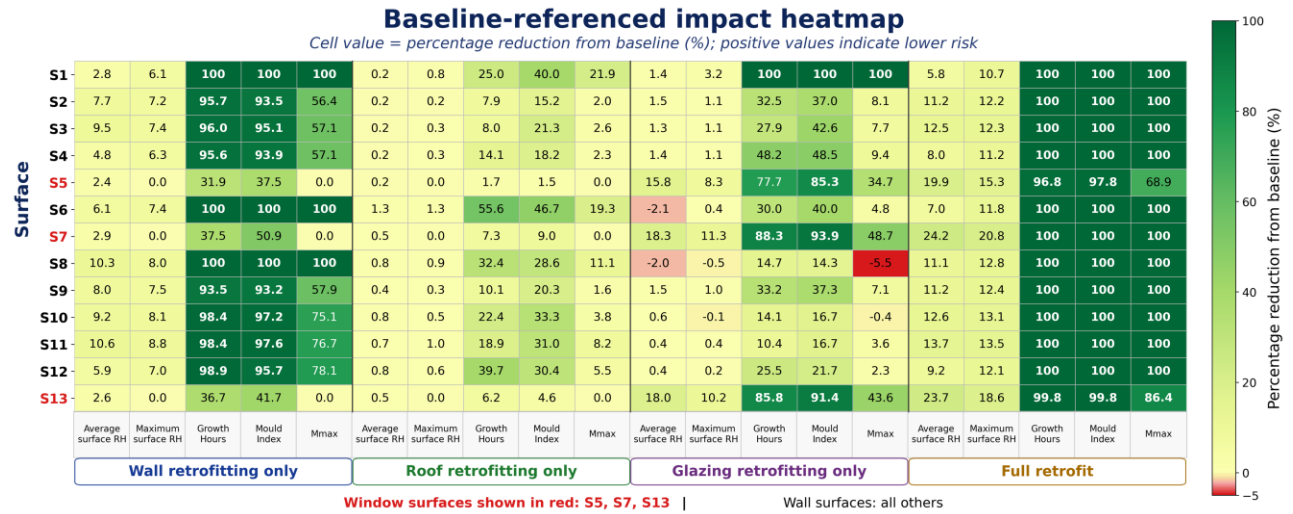


Figure 10. Percentage reductions in surface hygrothermal and mould indicators for each retrofit scenario.

While Figure 10 highlights the spatial distribution of these improvements, Figure 11 explains the underlying hygrothermal mechanisms. The retrofit scenarios follow distinct trajectories within the surface temperature-surface humidity space rather than producing uniform shifts. For the wall surfaces, the baseline condition (20.5 °C, 51.0%) moved only slightly following roof or glazing retrofits, whereas wall insulation increased the average ST to 22.2 °C while simultaneously reducing average SRH to 47.7%, decreasing the mean mould growth duration from 228 to only 7 h per surface. The full retrofit further shifted the wall surfaces to 22.7 °C and 46.3% SRH, eliminating mould growth entirely while reducing the average $M_{\max}$ to zero.

A different pathway was observed for the glazing-associated surfaces. The baseline state was characterized by substantially colder and wetter conditions (17.3 °C and 62.8% SRH), resulting in an average of 1,489 mould growth hours per surface and the maximum possible mould severity ($M_{\max} = 5.30$). Roof retrofitting produced almost no displacement from this state, while wall insulation primarily increased ST with only limited reduction in humidity, leaving 964 growth hours per surface and unchanged maximum severity. Glazing replacement, however, shifted the glazing surfaces much further towards warmer and drier conditions (19.8 °C and 53.6% SRH), reducing average growth duration to 243 h and lowering $M_{\max}$ to 3.06. The complete retrofit extended this trajectory to 20.5 °C and 49.7% SRH, leaving only 18 growth hours per surface and reducing the average $M_{\max}$ to 0.79.

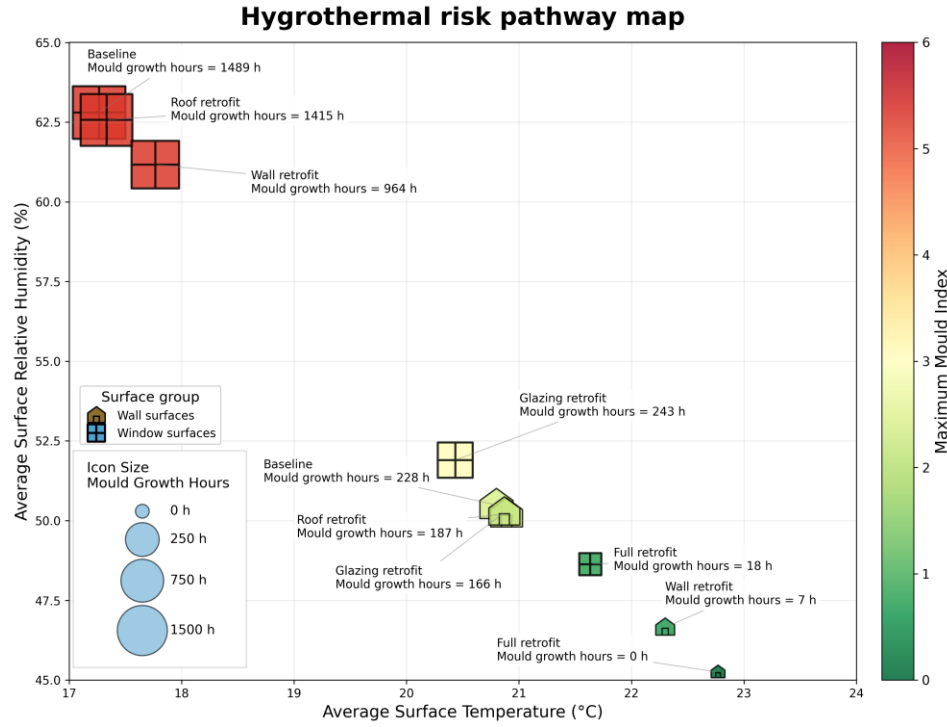


Figure 11. Hygrothermal risk pathway map for wall and window surface groups under the baseline and retrofit scenarios. Marker position represents the group-average ST and SRH, marker size represents mean mould growth hours per surface, and marker color represent the group-average M_{max} .

Taken together, the figures show that mould risk is controlled by the weakest untreated part of the envelope. Although the wall-only and glazing-only scenarios achieved substantial reductions, high residual M_{max} values persisted where vulnerable components remained. This means that average improvement alone is not enough to judge retrofit success. The key question is whether the intervention removes the surface that continues to govern peak mould severity. In this case, only the full retrofit addressed both the opaque walls and glazing, preventing risk from shifting to untreated locations and producing consistently low mould susceptibility across the enclosure.

Figure 12 compares the temporal evolution of the average mould index for window and wall surfaces under the wall-only, roof-only, glazing-only, and full-retrofit scenarios. Unlike Figures 4 and 5, it shows when mould growth occurred and how long it persisted. Seasonal peaks remained on untreated surface groups under the partial retrofits, whereas the full retrofit maintained the average mould index close to zero for both groups throughout the assessment period.

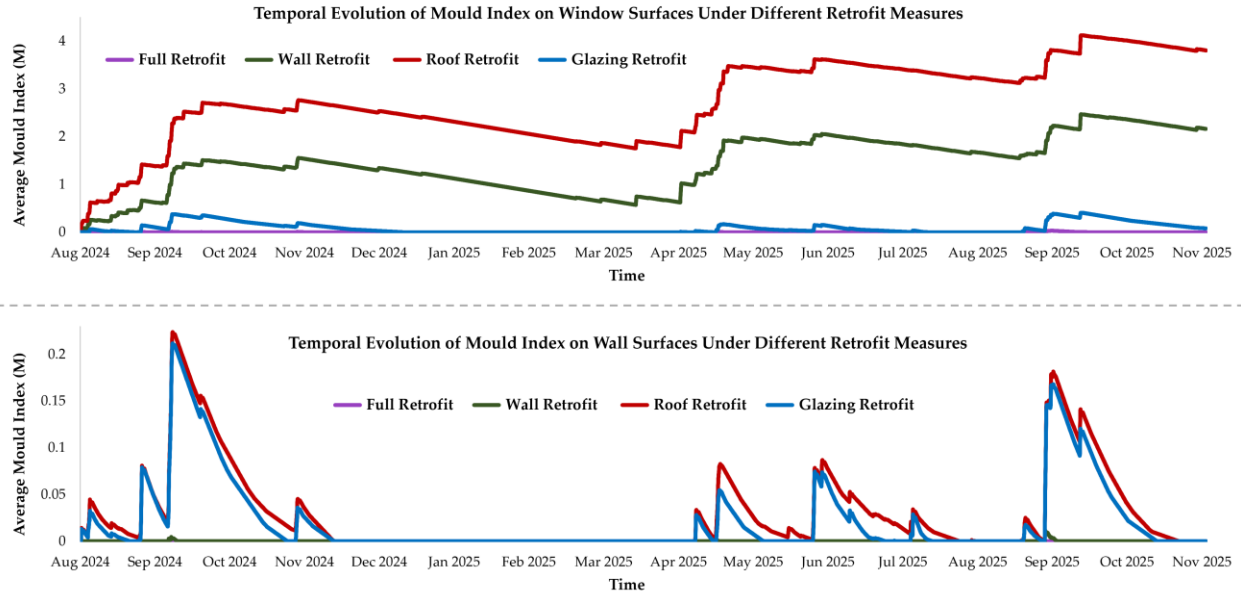


Figure 12. Temporal evolution of the average mould index across window and wall surface groups under the wall-only, roof-only, glazing-only, and full-retrofit scenarios.

4.4. Transfer Learning Across Building States

The TL results demonstrate that the proposed PIXGB model retains a high degree of predictive capability across multiple retrofit states, including the post-retrofit baseline, wall-only, roof-only, glazing-only, and full-retrofit scenarios at the 24 and 168 hours ahead forecasting horizons. Figure 13 summarizes the average testing RMSE across these target states, and Figure 14 provides representative time-series predictions for two selected cases.

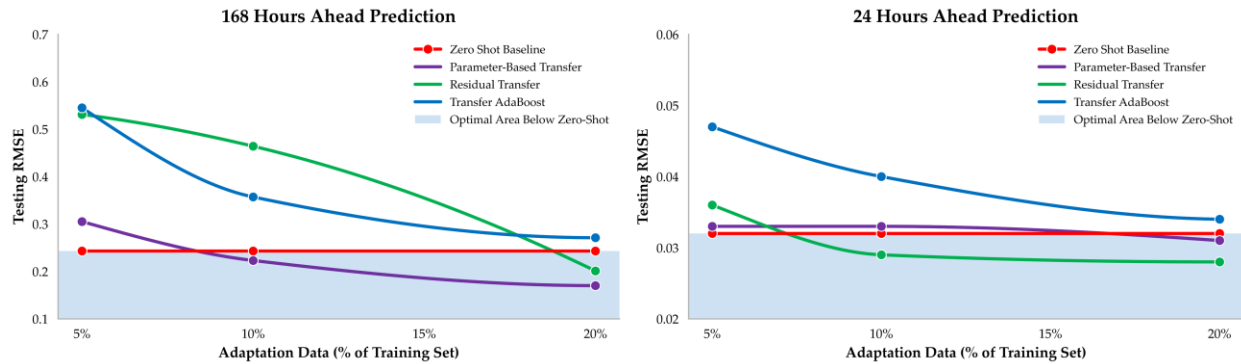


Figure 13. Testing RMSE of zero-shot and TL methods using 5%, 10%, and 20% adaptation data for the 168 h and 24 h prediction horizons.

A notable observation is the strong performance of the zero-shot model. With only 5% target data available, none of the TL strategies improved upon direct application of the source model, indicating that the physics-informed representation learned from the pre-retrofit building generalized well to previously unseen retrofit configurations. This behavior is expected with the design of PIXGB, whose feature space explicitly encodes the governing hygrothermal mechanisms through variables representing moisture accumulation, mould growth and decay rates, cumulative exposure, and temporal memory. Although the retrofit altered the magnitude of these variables by modifying the thermal behavior of the envelope, it did not alter the underlying physical processes governing mould development. Consequently, much of the knowledge learned in the source domain remained directly transferable without retraining.

As more target data became available, however, TL progressively exploited the remaining domain-specific differences. For the 168 h forecasting horizon, parameter-based transfer achieved the largest improvement, reducing the average RMSE from 0.243 for the zero-shot model to 0.170 with 20% adaptation data, corresponding to a 30.0% reduction in prediction error. Residual transfer also improved performance at this adaptation level, whereas Transfer AdaBoost remained consistently less effective. In contrast, the 24 h model favored residual transfer, which reduced RMSE from 0.032 to 0.028 using the same amount of adaptation data, representing a 12.5% improvement over zero-shot prediction. Parameter-based transfer provided only a marginal benefit for this shorter forecasting horizon.

The different adaptation behavior reflects the distinct error characteristics of the two prediction horizons. One-day-ahead short-term forecast are dominated by the current hygrothermal state and recent mould history, both of which are already well represented by the physics-informed feature engineering. Consequently, only a relatively small systematic error remains to be corrected, making residual learning the most efficient adaptation strategy in 24 h horizon. In contrast, in the one-week-ahead forecast, prediction errors accumulate over the 168 h horizon as deviations in thermal and moisture evolution compound over time. Updating the model parameters therefore becomes more effective than learning a residual correction, allowing the transferred model to better represent the modified dynamics of the retrofit scenarios.

The representative examples in Figure 14 illustrate these mechanisms. The upper panel shows the 168 h prediction for Surface S7 under the post-retrofit baseline scenario, where parameter-based transfer substantially reduces the persistent bias observed in the zero-shot prediction and more closely reproduces the measured seasonal evolution of the mould index. The lower panel presents the 24 h prediction for the same surface under the wall-retrofit scenario. Here, the zero-shot prediction already follows the observed response closely, leaving only a small residual error that is further reduced by residual transfer.

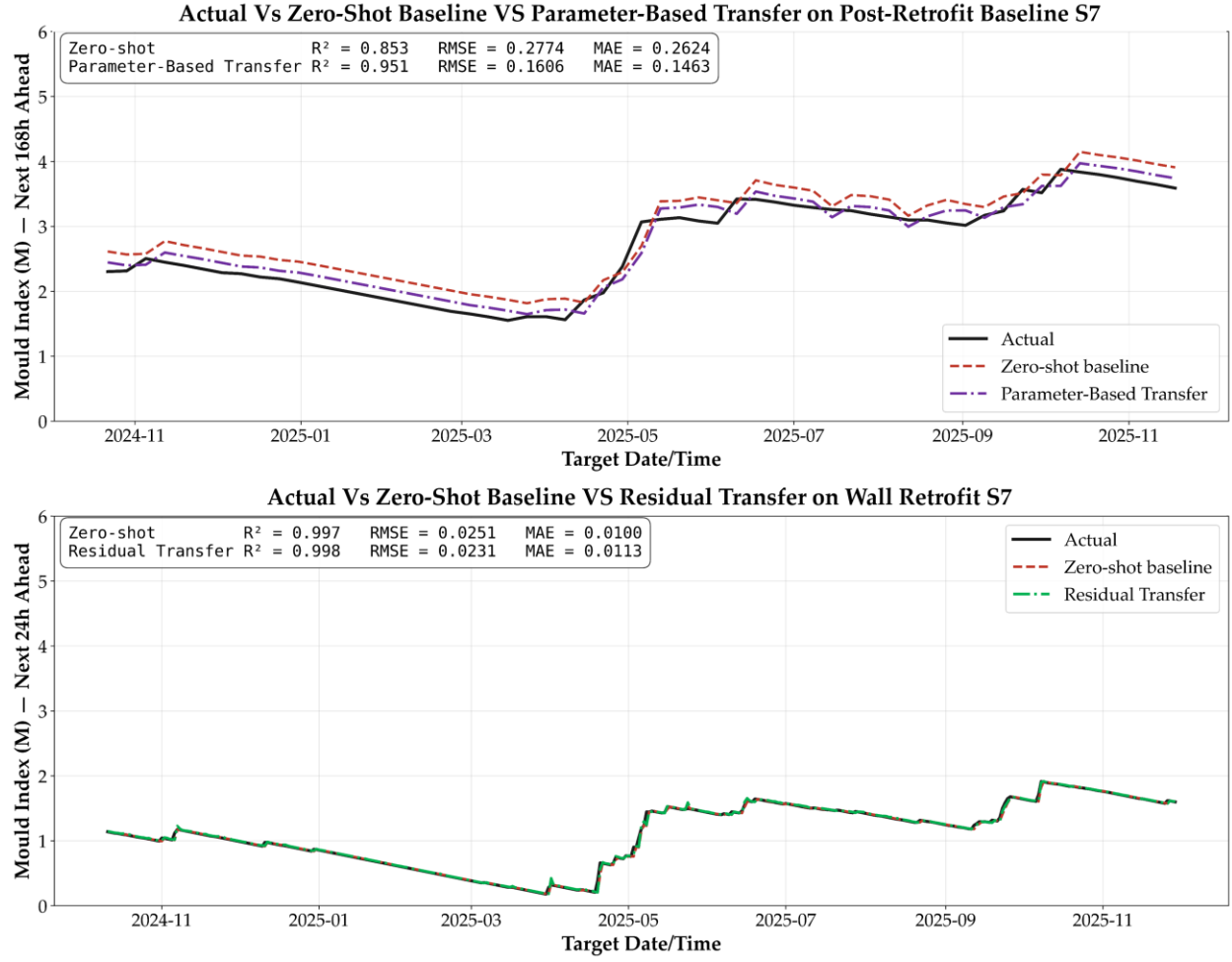


Figure 14. Representative actual, zero-shot, and transferred mould-index predictions at S7 for the 168 h post-retrofit baseline and 24 h wall-retrofit scenarios.

The results suggest that the principal value of TL is not to compensate for poor generalization of the source model, but to incrementally refine an already transferable physics-informed predictor as post-retrofit monitoring data become available. This is particularly attractive for digital twins operating in real buildings, where predictions can be deployed immediately after a retrofit using the existing model and progressively updated as new operational data accumulate, avoiding the need for complete model redevelopment after every major building intervention.

4.5. Physical Interpretation of the Transfer Learning Models

To examine whether TL preserved the physical reasoning of the original PIXGB model, SHAP (Shapley Additive Explanations) was applied to the best-performing transfer model for each forecasting horizon. Figure 15 presents the grouped SHAP values for the residual-transfer model at the 24 h horizon and the parameter-based transfer model at the 168 h horizon.

The transferred models remained dominated by physically meaningful predictors. For the one-day-ahead (24 h) model, Mould Process Physics remained the most influential feature group, followed by Hygrothermal Conditions, while Outdoor Environment, Indoor Air Quality, and Temporal Features contributed negligibly. This ranking is identical to that observed for the original PIXGB model, indicating that residual transfer refined the prediction without altering the underlying physical interpretation of the model.

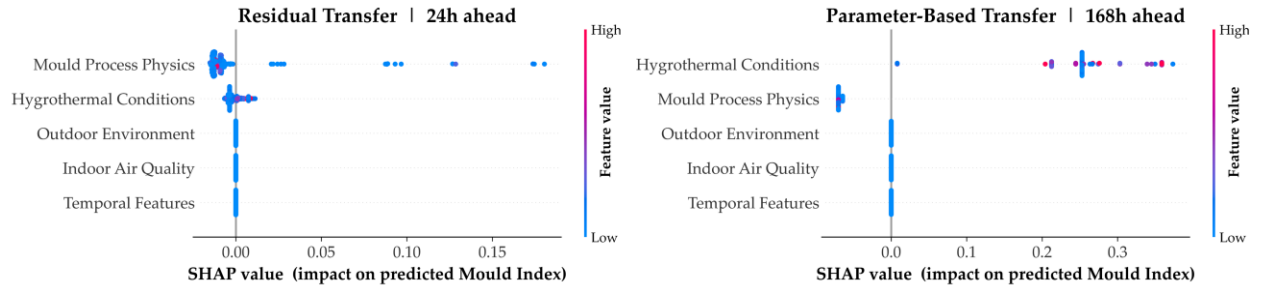


Figure 15. SHAP interpretation of the best-performing TL models for mould index prediction at the 24 h (residual transfer) and 168 h (parameter-based transfer) forecasting horizons. Feature groups are ranked according to their contribution to the predicted mould index.

A different feature hierarchy emerged for the one-week-ahead (168 h) model. Following parameter-based transfer, Hygrothermal Conditions became the dominant contributor, while Mould Process Physics became the second most important feature group. This shift is physically consistent with the longer prediction horizon. Short-term mould evolution is largely constrained by the current mould state and recent growth history, making process-based variables the strongest predictors. Over a one-week horizon, however, accumulated changes in ST and SRH increasingly govern future mould development. The transferred model therefore places greater emphasis on the evolving hygrothermal conditions that determine whether mould growth accelerates, stabilizes, or decays over the forecast period, while the importance of the current mould state naturally diminishes.

4.6. Limitations and Future Research Directions

Despite the strong performance of the proposed framework, the limitations should be acknowledged. First, mould susceptibility was quantified using the VTT mould index model, which was originally developed for wood-based materials and was later extended to other construction materials through sensitivity classes [61, 62]. Therefore, applying these generalized sensitivity classes to gypsum-based and other envelope surfaces provides a practical representation of mould susceptibility. More broadly, mould-growth models are derived from controlled experiments and empirical relationships between hygrothermal conditions and fungal development, and previous studies have reported differences between predicted mould risk and observed mould occurrence in real buildings [63–65]. The mould index reported in this study should therefore be interpreted as an indicator of hygrothermal mould susceptibility rather than direct evidence of biological colonization. Future research should validate and, where necessary, recalibrate the model using on-site inspection, visual assessment, and microbiological measurements under occupied building conditions.

Second, the proposed framework was evaluated using a single residential case-study building and one intensively monitored apartment. Although the integration of environmental monitoring, calibrated BEM, VTT-based mould assessment, and physics-informed prediction provides a transferable methodological structure, the quantitative findings remain specific to the investigated building and envelope configuration. Validation across additional building typologies, envelope systems, and climatic regions is therefore required to establish the broader applicability of the framework.

Third, the contribution of individual envelope interventions was evaluated using controlled counterfactual simulations rather than independently monitored physical retrofit stages. This enabled wall, roof, and glazing interventions to be compared under identical environmental and operational conditions and therefore isolated their relative effects more clearly than a direct pre- and post-retrofit comparison. Nevertheless, the resulting component-level effects remain dependent on the accuracy of the calibrated BEM and the assumptions adopted in the counterfactual scenarios. Future studies could strengthen this

evidence through staged retrofit monitoring or application across buildings where individual interventions are implemented separately.

5. Conclusion

This study extended a previously developed physics-informed AI-integrated Digital Twin framework to evaluate how building retrofit influences surface hygrothermal conditions, mould susceptibility, and predictive model performance as the physical building changes. The framework combined environmental monitoring, calibrated Building Energy Modeling, surface-level hygrothermal estimation, VTT-based mould assessment, and physics-informed machine learning. Using monitored pre- and post-retrofit conditions, the study first evaluated the overall effect of the completed retrofit on mould risk across the building envelope. Controlled counterfactual simulations were then used to isolate the contributions of wall insulation, roof insulation, glazing replacement, and the complete envelope retrofit under identical boundary conditions. Finally, TL strategies were evaluated to determine whether the existing pre-retrofit PIXGB models could retain and improve their predictive capability across modified building states using limited post-retrofit data.

The pre- and post-retrofit assessment demonstrated that the completed retrofit substantially improved the hygrothermal performance of the building envelope and reduced mould susceptibility across the investigated surfaces. Mean annualized mould-favorable exposure decreased by 96.6%, from 93.2 to 3.1 h per surface per year, with exposure eliminated at 11 of the 13 investigated locations. The improvement was particularly evident across the opaque envelope, where increased thermal resistance and improved thermal continuity reduced the colder and more humid surface conditions associated with the original wall assemblies. However, the results also showed that retrofit success cannot be assessed solely from average or whole-building improvements. Residual susceptibility remained at two glazing-related locations, S5 and S13, indicating that local junctions and thermally vulnerable details can remain important even after substantial overall improvements. The findings therefore demonstrate the importance of spatially resolved post-retrofit assessment for confirming not only whether mould risk has decreased, but also where residual risk remains.

The controlled retrofit scenarios further demonstrated that the effectiveness of individual interventions depended strongly on the envelope component governing local mould development. Wall insulation reduced mould-growth hours by more than 93% at most opaque wall surfaces, but provided only limited improvement at glazing-related hotspots. Conversely, glazing replacement reduced mould-growth hours at the window-associated surfaces by 77.7 to 88.3% and mould index by 85.3 to 93.9%, despite comparatively smaller changes in average SRH. Roof insulation produced only marginal improvements, indicating that roof heat transfer was not a dominant driver of mould risk in the investigated dwelling. Importantly, partial retrofit measures left elevated risk at untreated components, whereas the complete envelope retrofit addressed both wall and glazing vulnerabilities and maintained consistently low mould susceptibility across the enclosure. These findings show that mould-risk reduction is governed by the weakest remaining envelope component and that the relative value of retrofit measures should therefore be assessed according to their ability to eliminate critical local vulnerabilities.

The Transfer Learning analysis showed that the physics-informed PIXGB models retained substantial predictive capability after retrofit, allowing the pre-retrofit models to be applied directly to previously unseen building states with strong zero-shot performance. With only 5% target-domain data, none of the adaptation methods improved upon the zero-shot models, demonstrating the inherent transferability of the physics-informed representation. As additional post-retrofit data became available, however, targeted adaptation provided further improvements. At the 168 h horizon, parameter-based

transfer reduced average RMSE from 0.243 to 0.170 using 20% adaptation data, corresponding to a 30.0% reduction in error, while residual transfer reduced the 24 h RMSE from 0.032 to 0.028, a 12.5% improvement. SHAP analysis further showed that adaptation retained physically meaningful prediction mechanisms. Mould-process physics remained dominant for the 24 h forecast, whereas hygrothermal conditions became increasingly influential at the 168 h horizon, reflecting the greater importance of accumulated temperature and moisture dynamics over longer forecasting periods. The findings demonstrate that TL can progressively refine an already transferable physics-informed model without requiring complete redevelopment following changes to the physical building.

Overall, this study advances the application of Digital Twins for mould management from continuous monitoring and prediction toward retrofit verification, intervention prioritization, and adaptive post-retrofit prediction. The framework provides a practical means of determining whether a completed retrofit has effectively reduced mould susceptibility, identifying residual hotspots that may require further intervention, and distinguishing which envelope measures provide the greatest benefit under controlled conditions. For housing providers and facility managers, this capability can support more targeted allocation of retrofit resources, particularly where interventions must be prioritized or implemented in phases. At the same time, the TL framework enables an existing predictive Digital Twin to remain operational as the building evolves, allowing immediate post-retrofit prediction and progressive refinement as limited new monitoring data become available. By linking physical retrofit assessment with adaptive predictive capability, the proposed framework provides a pathway toward Digital Twins that can continuously evaluate, learn from, and respond to changes in the buildings they represent.

Abbreviations

Artificial Intelligence	AI	Mean Absolute Error	MAE
Actual Meteorological Year	AMY	Mean Bias Error	MBE
Air Relative Humidity	ARH	Outdoor Environmental Quality	OEQ
American Society of Heating, Refrigerating and Air-Conditioning Engineers	ASHRAE	Physics-Informed Extreme Gradient Boosting	PIXGB
Air Temperature	AT	Particulate Matter	PM
Building Energy Model	BEM	Coefficient of Determination	R ²
Building Information Modeling	BIM	Root Mean Square Error	RMSE
Coefficient of Performance	COP	SHapley Additive exPlanations	SHAP
Coefficient of Variation of the Root Mean Square Error	CVRMSE	Solar Heat Gain Coefficient	SHGC
Energy Efficiency Ratio	EER	Structured Query Language	SQL
Expanded Polystyrene	EPS	Surface Relative Humidity	SRH
Energy Recovery Ventilation	ERV	Transfer Learning	TL
Heating, Ventilation, and Air Conditioning	HVAC	Transfer AdaBoost for Regression	TrAdaBoost.R2
Indoor Environmental Quality	IEQ	Total Volatile Organic Compounds	TVOC
Internet of Things	IoT	Technical Research Centre of Finland	VTT

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